

Total No. of Questions : 8]

SEAT No. :

PB2513

[Total No. of Pages : 2

[6263]-399

B.E. (Artificial Intelligence and Data Science)

REINFORCEMENT LEARNING

(2019 Pattern) (Semester-VIII) (Elective-VI) (417533 D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q. 5 or Q. 6, Q. 7 or Q. 8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is dynamic programming, And how does it apply to solving Markov Decision Processes? **[9]**
- b) State the Banach Fixed point Theorem and its significance in dynamic programming. **[9]**

OR

- Q2)** a) Explain the contraction mapping property of Bellman expectation and optimality operators. **[9]**
- b) State and explain the principle of optimality in the context of MDPs. **[9]**

- Q3)** a) What are monte Carlo Methods, and how are they used in reinforcement learning. **[9]**
- b) Explain the idea behind per-decision Importance Sampling and its significance in off-policy learning. **[8]**

OR

- Q4)** a) What is the difference between On-policy and Off-policy learning in reinforcement learning. **[8]**
- b) What is Monte Carlo Tree Search (MCTS), and where is it commonly used? **[9]**

P.T.O.

- Q5)** a) Enlist the advantages and disadvantages of using model-based and model-free approaches in reinforcement learning. [9]
- b) Describe the Q-learning algorithm and its main components. [9]

OR

- Q6)** a) Discuss the double DQN algorithm and its advantages over traditional DQNs. [9]
- b) Explain the concept of Temporal difference (TD) learning in reinforcement learning [9]
- Q7)** a) How can an agent adapt when the model used for planning is inaccurate? [9]
- b) How do Rollout Algorithms help in approximating the value function and improving decision-making? [8]

OR

- Q8)** a) Explain the Dyna architecture and how it integrates planning, acting, and learning. [8]
- b) Discuss the advantages and limitations of using real-time Dynamic programming. [9]

